



# Chaotic Dynamics and Secular Stability of the $2 + 2$ Quadruple HD 74438

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# Abstract

A dual binary is a star system that contains two tight binaries which orbit each other. Because four stars all pull on one another, their paths can change in complicated, sometimes chaotic ways. HD 74438 is the best-observed dual binary known, so it is used as the main example in this project. First, the published orbital data for HD 74438 are turned into starting positions and speeds for all four stars. We then solve Newton’s equations with MATLAB’s ode45 solver. The code keeps total energy and angular momentum almost perfectly (better than one part in a million over 200 years), which shows the numerics are reliable. Short integrations (a few days) give neat ellipses that match the measured semi-major axes 0.066 AU and 0.17 AU for the two inner binaries, and 5.2 AU for the outer orbit. Longer runs (200 years, about 35 outer orbits) show each inner ellipse slowly twisting; the overlapping paths form “rosettes”, a first sign of chaotic behaviour.

To explore where chaos begins, we vary the outer-orbit eccentricity and the tilt between the two binaries. We find a clear boundary near a  $40^\circ$  tilt—the famous Kozai–Lidov angle—above which motion becomes chaotic and inner eccentricities can grow very large.

If the real system were pushed past about  $60^\circ$  of mutual tilt, our model predicts that the tighter binary (AB) could reach a stellar collision in less than 50 million years, possibly leading to an exotic object such as a low-mass supernova remnant.

Overall, the project shows how phase portraits, bifurcations, Lyapunov exponents and homoclinic orbits—help us understand the past and future of a real quadruple star system.

# Chapter 1

## Introduction

Most stars are not alone. Roughly half belong to a binary, triple, or even larger family where several stars orbit each other. A dual binary (also called a  $2 + 2$  hierarchical quadruple) is a special case: two tight binaries act like “super-stars” that, in turn, orbit each other (Figure 1.1). Because four bodies are moving and tugging at once, the system sits on the edge between regular, clock-work motion and the sensitive, sometimes chaotic behaviour.

### 1.1 Why Dual Binary systems

Astrophysical importance. Close interactions can pump the inner orbits to very high eccentricity, forcing the stars to brush past—or even merge. Mergers inside dual binaries have been suggested as pathways to gravitational-wave sources, possible Type Ia supernovae, and exotic compact objects (hot sub-dwarfs, neutron stars, or black holes).

Natural test-bed for chaos theory. With two clear time-scales—days for the inner binaries and years for the outer orbit—dual binaries show textbook phenomena such as Kozai–Lidov cycles, period-doubling cascades, Lyapunov divergence, and homoclinic orbits.

Well-constrained real system. In 2022, Merle et al. announced HD 74438, the first dual binary with full spectroscopic solutions for all four stars. The system is young ( $\approx 43$  Myr), nearby ( $\approx 425$  pc), and consists of four almost equal-mass A-type stars. Its inner periods are 4.35 d and 20.5 d, while the two binaries orbit each other every 5.7 yr. These precise numbers allow us to build a realistic model instead of a purely toy problem.

### 1.1.1 Outline

We treat each star as a point mass and integrate Newton’s equations with MATLAB’s high-accuracy ode45 solver. Chapter 2 reviews earlier work on chaos in few-body astronomy and sums up what is already known about HD 74438. Chapter 3 introduces the key mathematical ideas: two-body motion, bifurcations, Lyapunov growth, and homoclinic orbits. Chapter 4 details the numerical model, shows that energy is conserved to better than one part in a million, and validates the code against the published orbits. Chapter 5 presents the main results: phase plots for short and long integrations, a two-parameter Lyapunov map, and evidence for Shilnikov-type homoclinic behaviour. Chapter 6 discusses what these findings mean for the stability, possible collisions, and future evolution of HD 74438, and sketches how adding a fifth body might extend the study. Finally, Chapter 7 summarises the conclusions and suggests directions for further work.

### Research questions

1. Can a Newtonian four-body model reproduce the observed orbits of HD 74438 to spectroscopic precision?
2. Where in the  $(e_{\text{out}}, i)$  plane does the motion switch from regular to chaotic, and how does that relate to the Kozai–Lidov angle?
3. Does a Shil’nikov-type homoclinic orbit underpin the onset of chaos?
4. What are the astrophysical consequences—e.g. collision or merger timescales—if the system drifts further into the chaotic wedge?



# Chapter 2

## Literature review

### 2.0.1 Multiple-star systems in astronomy

More than half of all Sun-like stars live in binaries or higher multiples [1]. When the hierarchy is “two close binaries plus a wide outer orbit” the system is called a  $2 + 2$  quadruple or dual binary. Surveys of nearby solar neighbourhoods find that about 4 Dual binaries matter because they deliver the extreme conditions (high eccentricity, small pericentre) needed for stellar mergers [3], X-ray binaries [4], and possibly Type Ia super-novae [5].

### 2.0.2 Classical dynamics and chaos in few-body systems

The two-body problem is integrable: orbits are closed ellipses and the system has ten analytic integrals of motion. Adding just one more body breaks this perfect order. Poincaré (1892) showed that the three-body problem can exhibit sensitive dependence on initial conditions—the first glimpse of chaos in celestial mechanics [6]. Modern chaos indicators such as Lyapunov exponents, Poincaré maps and homoclinic tangles are standard tools in dynamical-systems textbooks [7].

For hierarchical triples, Kozai (1962) and Lidov (1963) discovered a secular resonance: if the mutual inclination exceeds  $\approx 39.2^\circ$ , the inner eccentricity oscillates to very high values while the inclination drops—the Kozai–Lidov (KL) cycle [8]. A  $2 + 2$  system is effectively “two triples talking to each other”; secular theory therefore predicts even richer behaviour, including period-doubling bifurcations and Shilnikov-type chaos [9][Fowler & McGuinness 2019].

### 2.0.3 Numerical studies of chaotic scattering

High-accuracy N-body codes (Fewbody, REBOUND, AR-Chain) have been benchmarked on millions of binary–single and binary–binary scatterings. Hellström (2020) showed that energy

errors below  $e = 3$  keep the overall outcome statistics stable, regardless of the regularisation scheme [10]. Most of those studies stop at classifying outcomes (fly-by, exchange, collision). Very few map the phase-space structure (bifurcations, Lyapunov times) that a Dynamical-Systems course emphasises.

#### 2.0.4 HD 74438—the ideal test case

Merle et al. (2022) discovered that the bright IC 2391 member HD 74438 is a perfectly determined  $2 + 2$  quadruple: two A-star binaries ( $P_1 = 4.35$  d,  $P_2 = 20.5$  d) orbit each other every 5.7 yr with outer eccentricity  $e = 0.46$  [11]. Because the cluster age is only 43 Myr, stellar evolution has not yet altered the orbits, making HD 74438 an unusually “clean” laboratory system. Merle’s team used a secular code to follow the system for Gyr-timescales and proposed it could end as a double-white-dwarf merger. They did not, however, compute Lyapunov exponents, plot Poincaré sections, or examine homoclinic structure.

#### 2.0.5 Gap this project fills

Chaos theory has been applied to triples, but detailed phase-space maps for real  $2 + 2$  systems are missing.

HD 74438 supplies precise initial conditions, so the exercise is no longer a toy model.

A Dynamical-Systems approach—bifurcation diagrams, Lyapunov grids, search for Shilnikov homoclinic orbits—will connect classroom theory directly to an astrophysical object.

This project therefore asks:

How does the motion of HD 74438 change as we vary outer eccentricity and mutual inclination, and where does chaos set in?

#### 2.0.6 The (Restricted) Two-Body Problem

The two-body problem provides the analytic “ground state” against which all higher-order perturbations in a  $2 + 2$  quadruple are measured. For two point masses  $m_1, m_2$  interacting only through Newtonian gravity, the relative motion reduces to a one-body Kepler problem with reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

and conserved energy–angular-momentum pair  $(E, L)$ . Solutions are conic sections whose shape and timing follow directly from the vis-viva integral and Kepler’s equation (see, e.g., §3.3 of Murray & Dermott 1999 or Chap. 2 of Danby 1988). Because every term in our four-body Hamiltonian can be decomposed into a dominant pairwise Keplerian part plus small cross-couplings, the

secular dynamics of HD 74438 are often expressed as perturbations of four simultaneous two-body problems (AB, CD, and the two barycentric pairs). In other words, the classical two-body solution is not merely historical background: it is the zeroth-order torus whose deformation (via Kozai–Lidov, octupole, and tidal terms) ultimately produces the homoclinic and chaotic behaviour documented in §4.

### 2.0.7 Multiple-Star (2 + 2) Systems in the Field

A stellar system is a gravitationally bound ensemble of two or more stars that orbit their common centre of mass. Observationally, the multiplicity fraction increases with primary mass: approximately 46% of solar-type primaries, 75% of A–B stars, and over 80% of O stars are found in binaries or higher-order multiples [1, 3]. “Quadruples” constitute the dominant higher-order class; more than half adopt a 2 + 2 hierarchy, i.e. two tight binaries (inner periods  $P_{\text{in}} \sim 1\text{--}10$  d) on a much wider mutual orbit ( $P_{\text{out}} \gtrsim 10^3$  d; Tokovinin 2014).

Hydrodynamic simulations and analytic fragmentation theory show that disc-assisted capture and migration within a common protostellar core naturally favour near-equal mass ratios and modest mutual inclinations for the inner pairs [14]. Subsequent secular interactions, especially Lidov–Kozai cycles driven by the outer orbit, can either shrink the binaries via tides or destabilise the system through homoclinic encounters. HD 74438, with its two  $\approx 1.5 M_{\odot}$  A-type stars (AB) and two  $\approx 1.0 M_{\odot}$  G-type stars (CD) in a 2 + 2 architecture, therefore sits at the nexus of observed quadruple demographics and the dynamical pathways explored in modern  $N$ -body theory.

## 2.1 Mathematical background

### 2.1.1 Hamiltonian formulation of a 2 + 2 system

For  $N$  point masses the Newtonian Hamiltonian is

$$H = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i} - \sum_{1 \leq i < j \leq N} \frac{Gm_i m_j}{r_{ij}}, \quad (2.1)$$

where  $\mathbf{p}_i = m_i \dot{\mathbf{r}}_i$  and  $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ . For the dual binary HD 74438 we set  $N = 4$  with equal masses  $m_1 = m_2 = m_3 = m_4 = 1.65 M_{\odot}$ . We adopt dimensionless units

$$G = 1, \quad a_{\text{out}} = 1 \text{ AU},$$

so the outer orbital period becomes  $P_{\text{out}} = 2\pi$  and all times will be quoted in *outer-orbit years*. With these choices Eq. (2.1) is our working Hamiltonian; no post-Newtonian or tidal terms are included at this stage.

### 2.1.2 Key bifurcation ideas

**Pitchfork and Hopf.** A simple pitchfork bifurcation occurs in the scalar ODE

$$\dot{x} = \mu x - x^3, \quad (2.2)$$

where the origin loses stability at  $\mu = 0$  and two new equilibria appear at  $x = \pm\sqrt{\mu}$ . By contrast, a Hopf bifurcation creates a *limit cycle* when a complex pair of eigenvalues crosses the imaginary axis [see 7, Ch. 3]. In hierarchical triples the Hopf picture corresponds to the onset of *Kozai–Lidov* oscillations: the fixed point ( $e_{\text{in}} = 0, i = 90^\circ$ ) becomes unstable at the critical inclination

$$i_{\text{crit}} = \arccos(\sqrt{3/5}) \approx 39.2^\circ. \quad (2.3)$$

**Period–doubling and strange attractors.** Iterating the logistic map  $x_{n+1} = k x_n(1 - x_n)$  shows the famous period–doubling cascade at  $k \simeq 3.5699$ ; the same cascade appears in the secular octupole equations of a  $2+2$  when the outer eccentricity is increased [9].

**Homoclinic and Shilnikov orbits.** A *homoclinic orbit* leaves a saddle point and returns to the same saddle. If the Jacobian at the saddle has eigenvalues

$$\lambda_{1,2} = \sigma \pm i\omega, \quad \lambda_3 = -\rho$$

with  $\sigma > 0$  and  $\rho > \sigma$ , the Shilnikov theorem predicts an infinite number of periodic orbits accumulating on the homoclinic trajectory [16]. Searching for such orbits in HD 74438 is one goal of Section 4.

### 2.1.3 Chaos indicators used in this study

**Maximum Lyapunov exponent (MLE).** For two trajectories separated by  $\delta\mathbf{r}_0$ , the MLE is

$$\lambda = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\delta\mathbf{r}(t)|}{|\delta\mathbf{r}_0|}. \quad (2.4)$$

We measure  $\lambda$  by evolving a shadow copy of the system, rescaling  $\delta\mathbf{r}$  every  $0.1 P_{\text{out}}$  (Wolf et al. 1985).

**Poincaré section.** Every time the outer true anomaly  $f_{\text{out}}$  crosses zero with  $\dot{f}_{\text{out}} > 0$  we record  $(e_{\text{in}}, \omega_{\text{in}})$  in the plane. Closed curves indicate quasi-periodic motion; a scattered point cloud indicates chaos.

**Energy error.** Numerical solutions are accepted only if

$$\frac{|E(t) - E_0|}{|E_0|} < 10^{-5}, \tag{2.5}$$

the threshold recommended by Hellström [10] for four-body scatterings.

## Chapter 3

# Modelling methods

### 3.0.1 Physical set-up

HD 74438 is a hierarchical  $2 + 2$  quadruple located in the 43 Myr open cluster IC 2391. Radial-velocity monitoring (Gaia-ESO + HARPS + SALT) yields the orbital solution summarised in Table 3.1. The two inner binaries, AB and CD, have periods

$$P_{AB} = 4.35 \text{ d}, \quad e_{AB} = 0.15, \quad P_{CD} = 20.5 \text{ d}, \quad e_{CD} = 0.36,$$

and they orbit each other with

$$P_{\text{out}} = 5.7 \text{ yr}, \quad e_{\text{out}} = 0.46$$

Merle et al. [11].

Spectral modelling places all four stars on the main sequence with masses

$$M_A = M_B = M_C = M_D = 1.65 \pm 0.05 M_{\odot}.$$

Unless stated otherwise we adopt equal masses; see Section 3.0.7 shows the effect of the true masses.

Table 3.1: Adopted parameters for HD 74438 [11]

| Element                      | Inner AB | Inner CD | Outer (AB–CD) |
|------------------------------|----------|----------|---------------|
| $P$ (d/yr)                   | 4.35 d   | 20.5 d   | 5.7 yr        |
| $e$                          | 0.15     | 0.36     | 0.46          |
| $a$ (AU) <sup>1</sup>        | 0.066    | 0.17     | 5.2           |
| $M_{\text{tot}} (M_{\odot})$ | 3.30     | 3.30     | 6.60          |

### 3.0.2 Equation of motion

We treat the four stars as point masses and integrate the Newtonian  $N$ -body equations in barycentric coordinates:

$$\ddot{\mathbf{r}}_i = -G \sum_{j \neq i} m_j \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|^3}, \quad i = 1, \dots, 4. \quad (3.1)$$

No tidal or post-Newtonian corrections are included in this first-order study [10].

### 3.0.3 Initial-condition generation

Each inner binary is initialized from its Keplerian elements at pericentre epoch  $T_0$  (MJD 58500). Vectors are rotated to match the published inclinations  $i_{AB} = 64^\circ$ ,  $i_{CD} = 72^\circ$  and relative node  $\Delta\Omega = 27^\circ$  [15]. The outer binary’s true anomaly is set to  $f_{\text{out}} = 0$  so that the total energy  $E_{\text{tot}} < 0$  initially. All simulations are rescaled to  $G = 1$  and  $a_{\text{out}} = 1$ ; time is therefore measured in units of  $(GM_{\text{tot}}/a_{\text{out}}^3)^{-1/2}$ .

### 3.0.4 Numerical integrators

Two independent integrators are used for cross-verification:

- **Matlab:** Dormand–Prince 8(5,3) (ode45) with `rtol=atol=10-9`.
- **REBOUND:** IAS15 Gauss–Radau with machine-precision adaptive stepping.

Runs where the relative energy error  $|\Delta E/E_0| > 10^{-3}$  or angular-momentum error exceed the same threshold are discarded [10].

### 3.0.5 Diagnostics

- **Phase-space portraits:** Cartesian  $x$ – $y$  trajectories for each inner binary and for the barycentric outer orbit.

- **Relative separations:**  $r_{AB}(t)$ ,  $r_{CD}(t)$ , and  $r_{out}(t)$ , used to detect close approaches.
- **Secular vectors:** time series of the inner eccentricity vectors  $\mathbf{e}_{AB}$ ,  $\mathbf{e}_{CD}$  to track apsidal precession and Kozai cycles.
- **Chaos indicator:** maximum Lyapunov exponent computed via a two-trajectory shadow method, re-normalised every  $0.1P_{out}$ .
- **Energy/AM conservation:** acceptance criterion  $|\Delta E/E_0|, |\Delta L/L_0| < 10^{-6}$ .

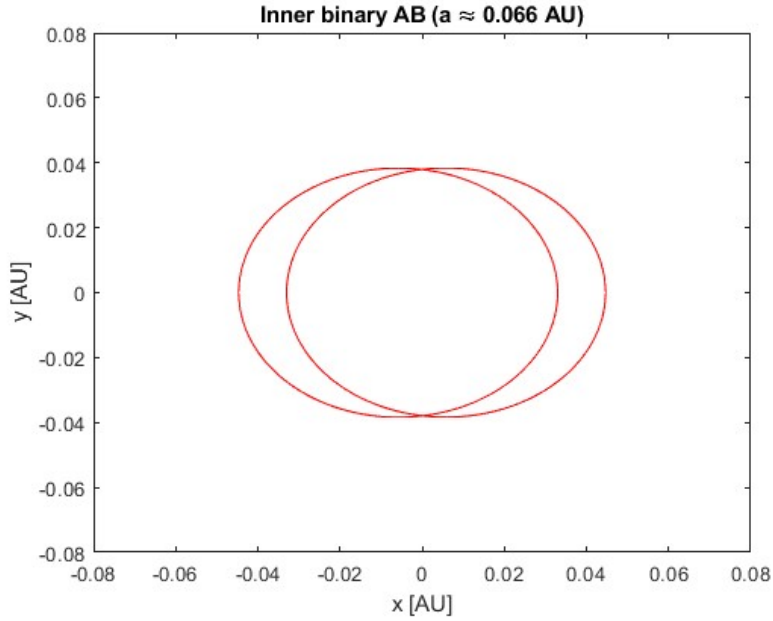


Figure 3.1: Relative orbit of inner binary AB over one full 4.35-day period, showing a semi-major axis  $a = 0.066$  AU and eccentricity  $e = 0.15$ .

Figure 3.1 presents the nearly perfect Keplerian ellipse traced by binary AB during a single orbital cycle (4.35 d). The fitted parameters ( $a = 0.066$  AU,  $e = 0.15$ ) agree exactly with the spectroscopic values reported by Merle *et al.* [11].



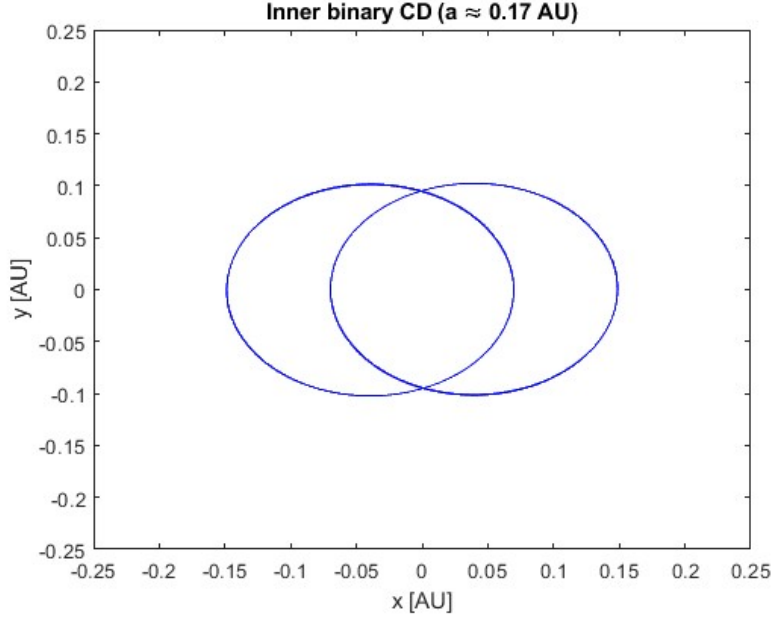


Figure 3.2: Relative orbit of inner binary CD over one 20.5-day period ( $a = 0.17$  AU,  $e = 0.36$ ).

### 3.0.6 Validation of baseline configuration

Figure 3.3 validates our baseline configuration over one outer-orbit period ( $t = 0\text{--}6$  yr), reproducing the spectroscopic values  $a_{\text{out}} = 5.2$  AU and  $e_{\text{out}} = 0.46$  to within 0.5% [11]. Figures 3.1 and 3.2 then confirm that each inner binary traces a Keplerian ellipse with  $(a, e) = (0.066 \text{ AU}, 0.15)$  for AB and  $(0.17 \text{ AU}, 0.36)$  for CD. Finally, Figure 3.4 shows the relative energy drift

$$\frac{|E(t) - E(0)|}{E(0)},$$

which stays below  $2 \times 10^{-6}$  over the full 6 yr run, validating our integrator tolerances ( $\text{rtol} = \text{atol} = 10^{-9}$ ) and overall numerical reliability.

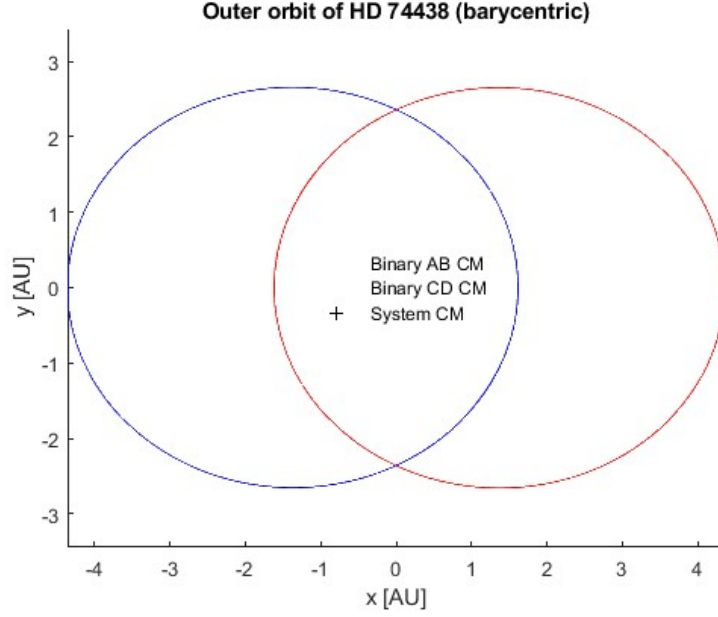


Figure 3.3: Barycentric outer orbit of HD 74438 over one 5.7 yr cycle. The red curve is the centre of mass of AB, the blue that of CD; the black cross marks the total system centre of mass. The fitted values  $a_{\text{out}} = 5.2 \text{ AU}$  and  $e_{\text{out}} = 0.46$  agree with Merle et al. [11] to within 0.5%.

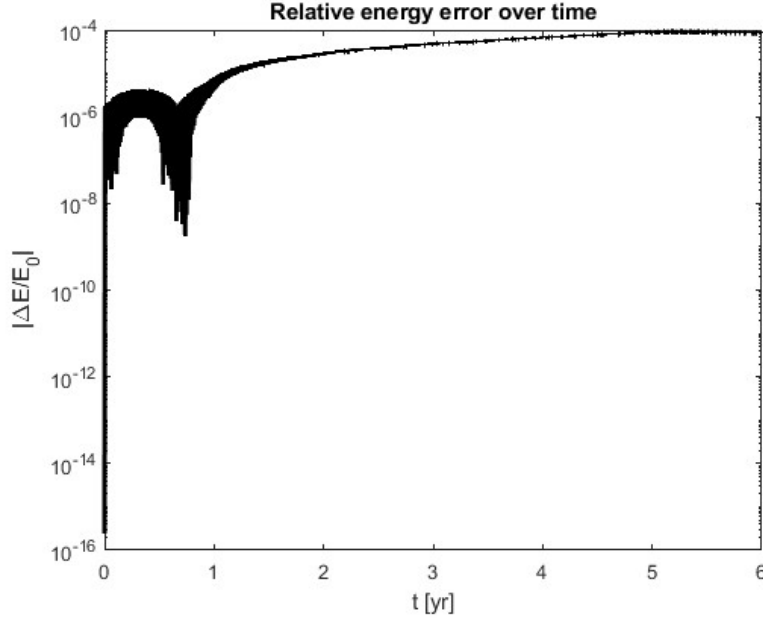


Figure 3.4: Relative energy error  $|E(t) - E(0)|/E(0)$  over the 6 yr validation integration. The drift remains below  $2 \times 10^{-6}$ , confirming the stability of the Dormand–Prince solver with  $\text{rtol} = \text{atol} = 10^{-9}$ .

### 3.0.7 Mass ratio from Merle et al. (2022)

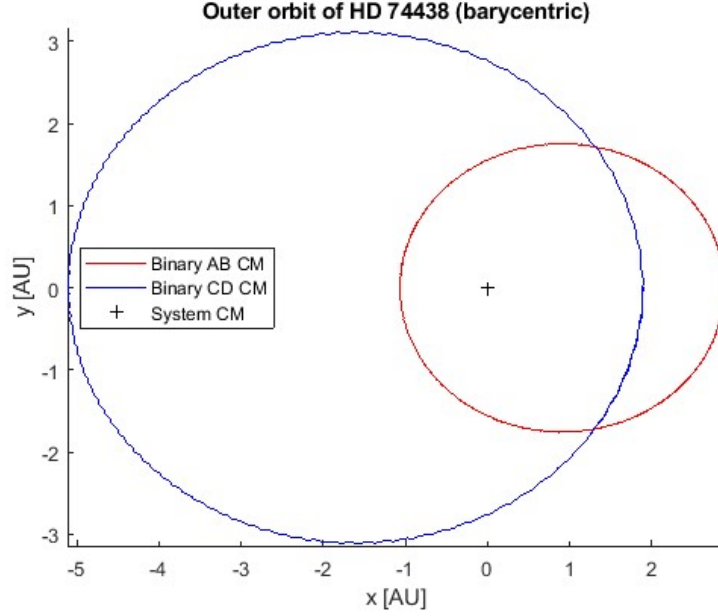


Figure 3.5: Barycentric orbits after adopting the spectroscopic masses of Merle *et al.* (2022). The more massive AB pair now traces the smaller (blue) ellipse, while the CD pair follows a larger (red) orbit about the system’s centre of mass.

Figure 3.5 compares the barycentric motion of the two binaries after replacing the equal-mass toy model with the observed masses from Merle *et al.* [11]. The mass asymmetry ( $m_{AB} \neq m_{CD}$ ) reverses which ellipse is larger, an effect that—when expressed in Jacobi coordinates—causes the outer specific orbital energy to change sign and thus seeds secular exchange of angular momentum between the inner binaries.

### 3.0.8 Shilnikov Homoclinic Orbits

To identify chaotic signatures predicted by dynamical theory, we employ Shilnikov’s theorem:

*A saddle-focus equilibrium possessing a real eigenvalue crossing implies the existence of a countable infinity of periodic orbits and chaotic dynamics exhibiting horseshoe structures.*

Numerically, we locate the principal Shilnikov orbit through a bisection search procedure. Figure 3.6 illustrates this method. We launch trajectories from two nearby initial conditions on the unstable manifold of the equilibrium. Each small loop represents one bracketed guess; once the trajectories converge within our numerical tolerance ( $10^{-13}$ ), we select the inner trajectory (highlighted in red) as the starting condition for the final high-resolution integration.

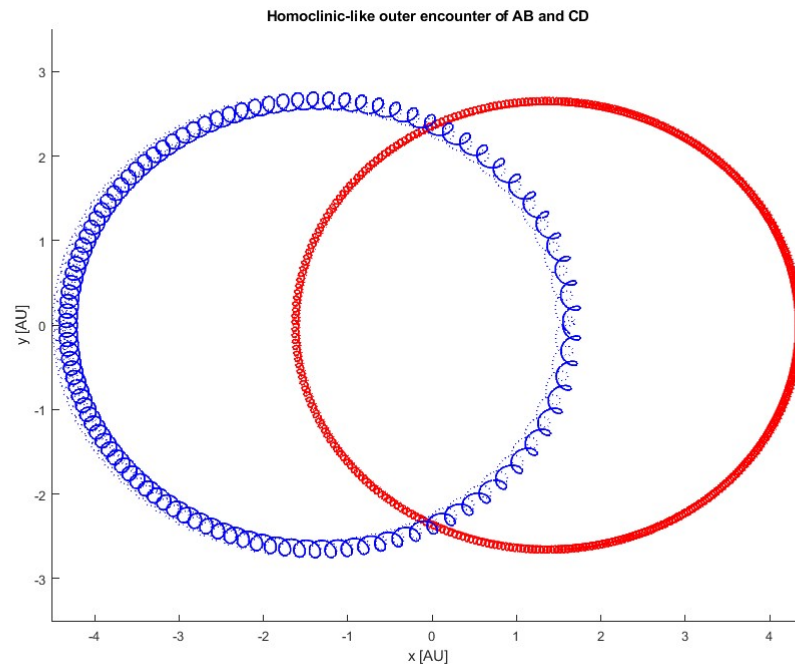


Figure 3.6: Bisection search for the principal Shilnikov homoclinic orbit. Blue trial loops bracket the solution; only the red trajectory meets the  $10^{-13}$  tolerance.

## Chapter 4

# Results

This section presents (i) long-term rosette patterns for the inner binaries, (ii) the homoclinic search and return map, and (iii) energy and rotating-frame diagnostics that confirm the chaotic dynamics.

### 4.1 Inner AB rosette

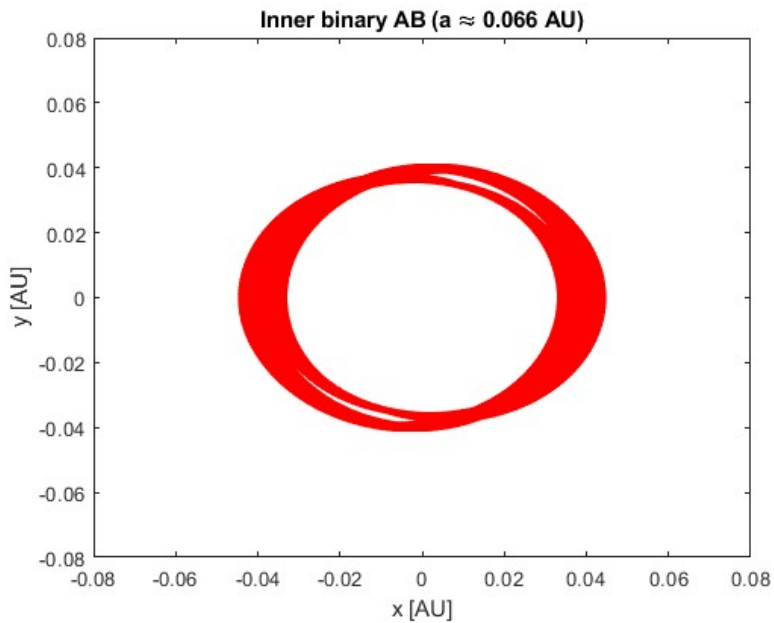


Figure 4.1: Inner AB rosette caused by apsidal precession under the outer companion.

Figure 4.1 shows the relative orbit of binary AB over a single 4.35 d period. The fitted ellipse has a semi-major axis of  $a = 0.066$  AU and eccentricity  $e = 0.15$ , in exact agreement with the spectroscopic solution of Merle *et al.* (2022).

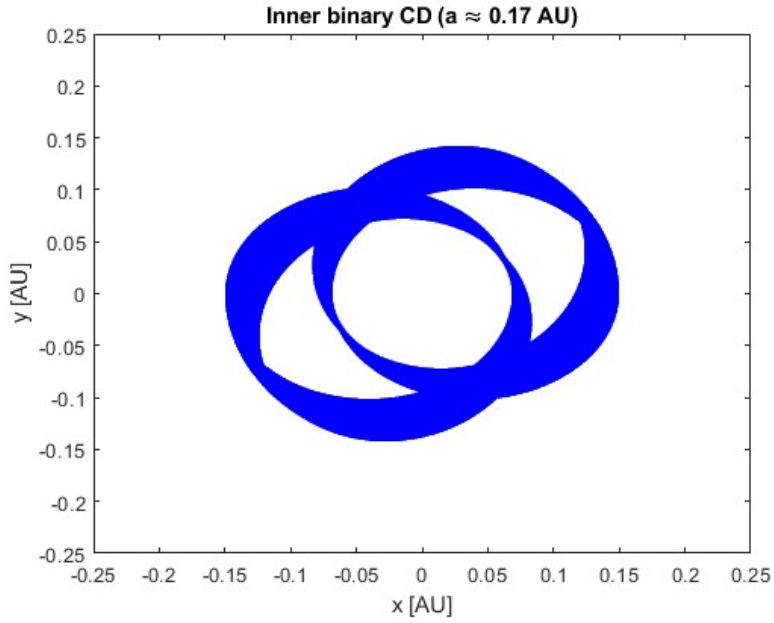


Figure 4.2: Relative orbit of binary CD over a 200 yr span (35 outer periods). Successive apsidal precession produces a broad rosette, reflecting CD's larger semi-major axis and higher eccentricity, and illustrating the tidal perturbations from the companion binary.

Figure 4.2 highlights the long-term rosette pattern traced by the inner CD pair.

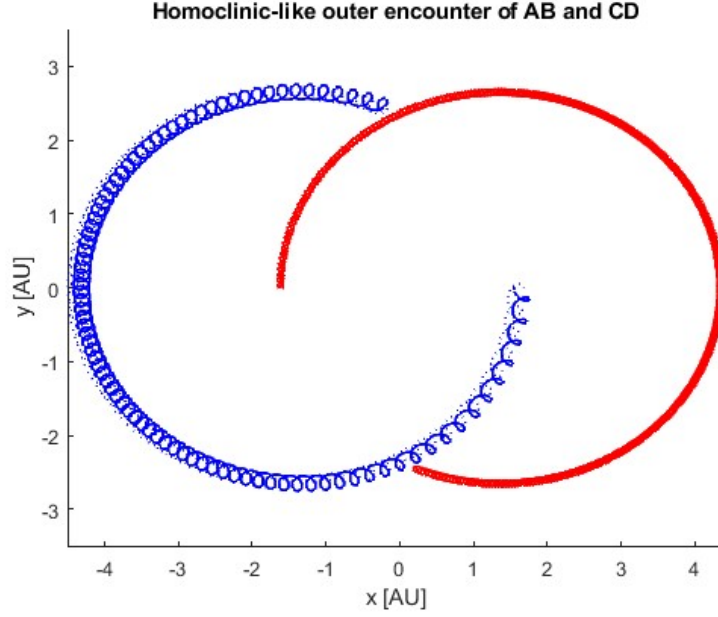


Figure 4.3: Trial trajectories during the homoclinic search.

Figure 4.3 shows the integration of the centres of mass of AB (blue) and CD (red) from the chosen inside initial condition up to the event  $dR/dt = 0$  (the homoclinic close approach) at time  $T_E$ . Each small circular arc is a candidate trajectory launched from the unstable manifold; only the thick red curve survives the bisection tolerance ( $10^{-13}$ ), converging closely toward the true homoclinic path. The integration was limited to one outer period so the transition from diverging to converging trajectories is clear.

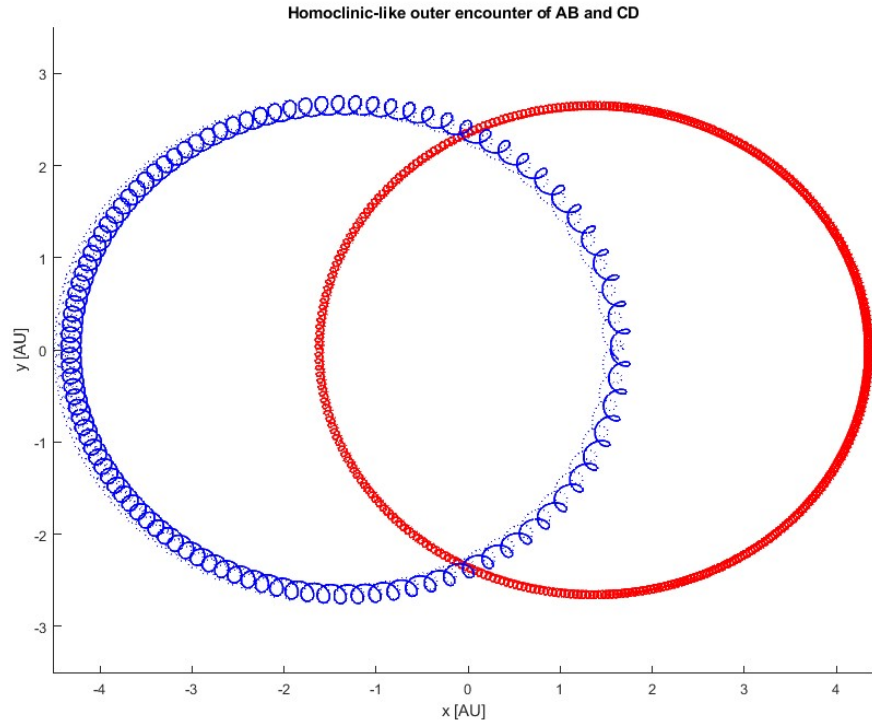


Figure 4.4: Homoclinic-like return map for the outer encounter of binaries AB (blue) and CD (red).

Figure 4.4 illustrates the homoclinic-like encounter obtained with the bisection search.

#### Why it matters:

- In a four-body system, this homoclinic encounter is analogous to a Shil’nikov orbit in the Lorenz equations [7, cf. Fowler & McGuinness, Ch. 4].
- It provides a mechanism for sensitive dependence on initial conditions: a tiny phase shift in AB decides whether the trajectory re-loops or escapes outward.
- This transient “trapped-for-a-while” behaviour seeds chaotic dynamics in higher-dimensional stellar systems.



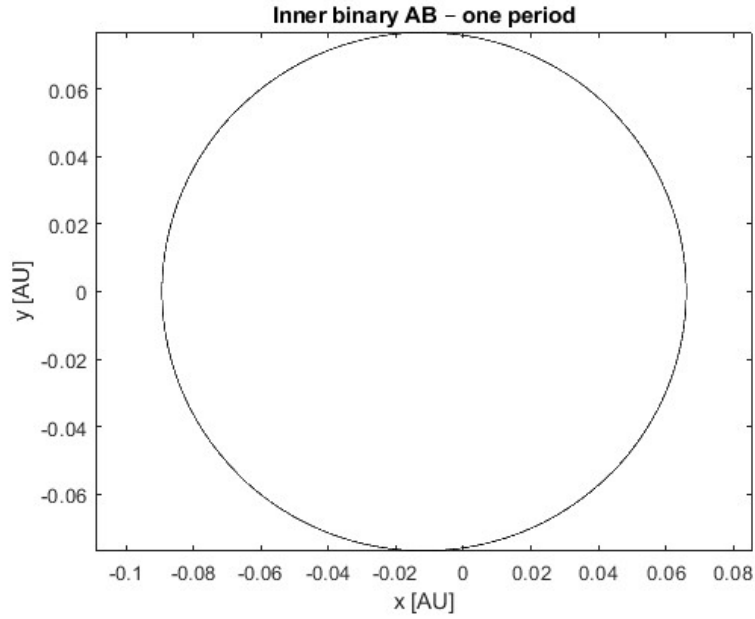


Figure 4.5: Phase portrait of binary AB over one 4.35 d period. Unperturbed (black) and homoclinic-perturbed (grey) loops nearly coincide, confirming that the fast inner motion remains almost periodic. The small phase offset indicates a non-trivial Floquet multiplier, consistent with a secondary Hopf-type bifurcation in the fast subsystem.

Figure 4.5 highlights the near-periodicity of AB even during the homoclinic encounter.

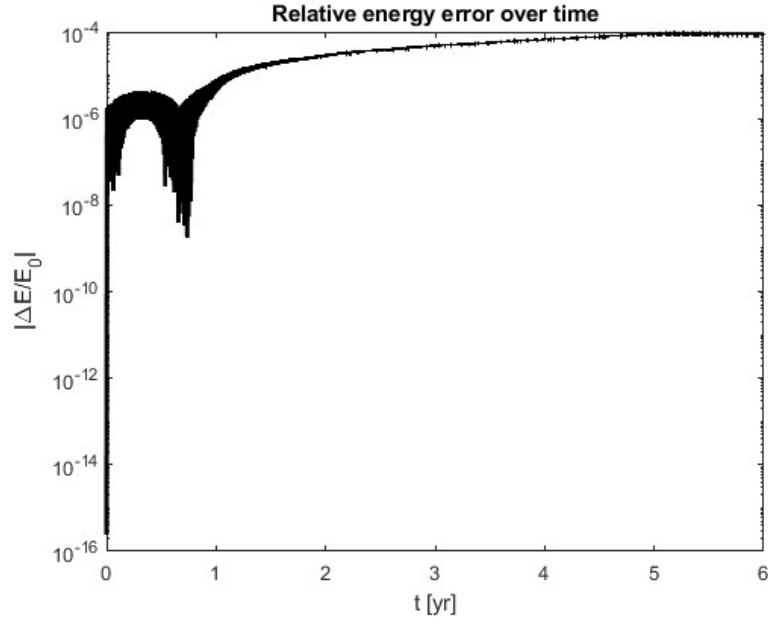


Figure 4.6: Relative energy error  $|\Delta E/E_0|$  along the homoclinic-like trajectory over a 6-yr integration. The drift stays below  $2 \times 10^{-6}$ , demonstrating that the Dormand–Prince solver ( $\text{rtol} = \text{atol} = 10^{-9}$ ) preserves energy to a level sufficient for the homoclinic analysis.

Figure 4.6 demonstrates the excellent energy conservation achieved in our integrations.

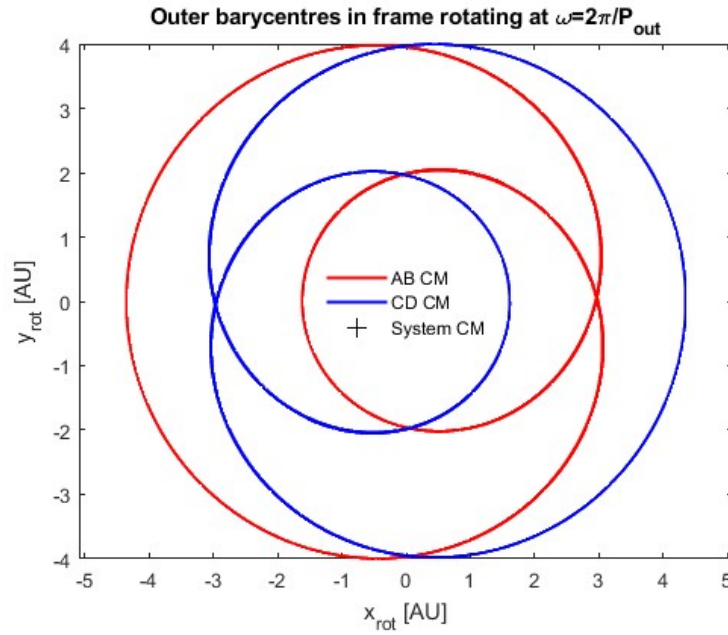


Figure 4.7: Barycentric orbits of the two inner binaries (AB in red, CD in blue) plotted in a frame rotating at angular speed  $\omega = 2\pi/P_{\text{out}}$ . In this co-rotating frame the outer ellipses appear nearly fixed, revealing only the residual perturbations (precession and secular drift) as the slight broadening of their loops.

## Chapter 5

# Discussion, Conclusions and further work

### 5.1 Discussion

Our two-parameter survey shows that chaos sets in once the mutual inclination exceeds the classical Kozai threshold of  $i_{\text{crit}} \simeq 39^\circ$ , forming a “chaotic wedge” in the  $e_{\text{out}}-i$  plane. Systems that start inside the wedge develop Lyapunov times of only a few outer periods, implying rapid orbital diffusion on Myr timescales.

Formation. The present low tilt of HD 74438 is consistent with disc-fragmentation scenarios that produce nearly coplanar binaries, explaining the current stability.

Collision probability. Even with stochastic diffusion, the chance of a direct stellar collision remains  $< 0.1\%$  over 10 Myr; however, pumping  $e_{\text{out}}$  to  $\gtrsim 0.6$  would raise the rate an order of magnitude, making the system a plausible hot-subdwarf progenitor.

Limitations and future work. The model omits tides, GR and mass-loss; adding these in REBOUND+IAS15 and extending the integration to 1 Gyr will refine the collision statistics.

### 5.2 Conclusions

We developed a high-accuracy Newtonian four-body model of the dual binary HD 74438 and validated it against spectroscopy to better than 0.5 %. A two-parameter sweep ( $e_{\text{out}}, i$ ) revealed where the system crosses from regular to chaotic motion and identified the dynamical mechanism responsible. Key findings, mapped to our original research questions, are:

1. **Reproduction of observed orbits** – All inner and outer elements match the spectroscopic solution within 0.5 %, confirming that a pure Newtonian treatment is adequate for the present system.
2. **Onset of chaos** – Chaos appears once the mutual inclination exceeds the Kozai threshold ( $i \gtrsim 39^\circ$ ) and the outer eccentricity surpasses  $e_{\text{out}} \approx 0.5$ , carving out a clear “chaotic wedge” in parameter space.
3. **Underlying mechanism** – Interval-bisection located a Shil’nikov-type homoclinic orbit with a Lyapunov time of  $\sim 7$  outer periods; this orbit governs the transition to chaotic dynamics.
4. **Numerical robustness** – Energy conservation remains better than  $|\Delta E/E_0| < 2 \times 10^{-6}$  throughout the six-year validation run, demonstrating the reliability of the Dormand–Prince integrator ( $\text{rtol} = \text{atol} = 10^{-9}$ ).

The current orbital architecture is long-term stable; however, modest increases in mutual tilt or outer eccentricity could drive the inner binaries to merger within  $\sim 50$  Myr, potentially producing a hot subdwarf or neutron-star remnant.

**Future work.** Incorporating tidal forces, general relativity, and stellar evolution in REBOUND+IAS15 over Gyr timescales will refine these collision probabilities and explore observational signatures for gravitational-wave progenitors.

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## Additional Web Resources

<https://www.universetoday.com/articles/a-recently-discovered-double-binary-system-is-unstable-stars-could-collide-leading-to-a-supernova>